CHAPTER 5

THE CASE OF INDIAN RAILWAYS[5]

5.1 INTRODUCTION

In India, the seeds for the development of Indian Railways (IR) were sown from the mere idea to connect the suburbs of Bombay (now Mumbai) in 1843. Formally inaugurated a decade later, it took one more year for the first passenger train to steam out of the Howrah station. During the next three decades, Indian Railways had a route mileage of 9000 miles (almost one-fourth of today's route mileage). From a mere idea, the IR, considered to be most important mode of transportation in the country, has come a long way to become the largest rail network in Asia and the world's second largest under one management (Railways, 2015a). Over such a wide and complex network, IR operates around 7000 passenger trains and 4000 freight trains.

To operationalize this, IR has 10,773 locomotives, 51,798 passenger carriages, 2,54,006 freight wagons running over a network of 7,137 stations. To park the idle equipment, IR has 300 Yards, 2300 Good sheds and 700 Repair shops. The entire operation is made possible with the help of 1.54 million work force (Railways, 2015b). To ensure smooth and efficient working of such a wide network, IR has been divided in to 16 zones with each zone further divided in to divisions.

[5] A part of this Chapter has been published in IIM Kozhikode Society and Management Review

Despite the steady increase in operating revenues for the IR, the operating expenditure has been growing and at a rate more than the operating revenue (Railways, 2015b). In the passenger segment, IR claims to recover only 57% of the cost incurred. On the assets front, CAG (2013), audit report based on the audit conducted on Indian Railways reported that 90 per cent of the loco sheds carried excess holdings of locomotives. As a result, maintenance schedule could not be carried out in 21 per cent of diesel locomotives which were due for maintenance.

This indicates the requirement of more locomotive sheds which need to be strategically placed so as to minimize the overall cost. Also, there were substantial delays in sending locomotives for Periodic Overhaul (POH), with delays ranging up to 360 days. This could be a result of not having the requisite number of locomotives to run the operation or inefficient utilization of the locomotives.

Further, 65 per cent of the overhauled locomotives registered failure within 180 days of POH. 17-20 per cent of them failed within one month of POH indicating poor standards of POH. A test check of 28 loco sheds revealed that 11,626 locomotives were given for 'out of course repairs' during a nine months period and resulted in 15810.64 ineffective engine days and loss of earning capacity of Rs. 2.81 billion. There was extra detention for completing scheduled maintenances and POH resulting in loss of earning capacity of Rs. 2.10 billion.

The excess detention of locomotives in yards (more than a day) prior to POH and after POH as well as delays in transfer of dead locomotives for repairs to the loco sheds resulted in estimated earnings loss of Rs. 2.41 billion. The expenditure on unscheduled repairs was estimated at about Rs. 810 million. All these problems indicate poor maintenance and overhauls of locomotives arising primarily out of capacity constraints.

While, according to Railways (2015a), there has been an increase in the participation of both private and public sectors in an attempt to develop the infrastructure, to ensure sustained participation, IR needs to reduce its operating expenses and increase the profit margin. The above-mentioned problem is only compounded by the social obligations and care for weaker sections that IR as a public-sector entity, has to cater to. Concessional benefits to senior citizens, students, disabled persons etc. cannot be done away with. However, the operational efficiency can be increased to a great extent to achieve this goal. There is a great scope to this end in terms of optimizing the locomotive assignments and identifying locations for locomotive sheds.

5.2 STUDIES PERTAINING TO INDIAN RAILWAYS

Railways management has been a topic of interest among researchers over the past few decades. With a railway network as vast as India's, researchers have attempted to address the issues faced by Indian railways as well. In this section, we review the published literature with specific reference to Indian Railways.

Ramani & Raghuram (1979) attempt to address the problem of shortage of coaches for passenger services in Indian Railways. They develop an indicator to monitor operations and predict the efficiency of any proposed modification and use the same indicator to analyse utilization of other transportation units also. Ramani & Mandal

(1992) design, develop and implement an optimization-based decision support system (DSS) for the operational planning of Indian Railways. For the DSS, they generate an optimization model to identify optimal train links and a flexible model to examine implementation feasibility. A couple of zonal railways have implemented the system and reported benefits.

Kumar, Vrat, & Sushil (1994) conduct a simulation study on the repair inventory system of diesel locomotives in Indian Railways. They develop a system dynamics framework and conduct an empirical study over a network of ten diesel locomotive sheds and one remanufacturing workshop. They provide a decision support system to help managers to adopt better spares management policies.

Raghuram & Babu (1999) focus on the financial requirements of Indian Railways and suggest some alternate modes of financing railway projects. They first examine the past investment patterns based on the actual investments made by railways. They also study the various investment modes available as options and their pros and cons.

Kulshreshtha, Nag, & Kulshrestha (2001) attempt to estimate freight transport demand. They use a multivariate cointegrating vector auto regressive model to arrive at the estimate. To ascertain the cointegration rank and estimates, they use Johansen’s maximum likelihood approach. They perform the analysis over two sub-samples as well as the full sample to check the stability of cointegrating relationships. They claim that the demand system in the long run is stable as it achieves equilibrium in a period of 3 years after facing a system-wide shock.

Pucher, Korattyswaroopam, & Ittyerah (2004) study the crisis of public transport arising out of overwhelming demand being met with limited resources. They study the problems and challenges faced by public transport systems in general and suggest privatization as one of the ways to overcome them. Agarwal (2008) study customer satisfaction in public transportation in the context of Indian Railways. The author uses the survey method to conduct the study and performs a factor and regression analysis on a sample of 500 data points. The study reports employee behaviour to be the most important factor impacting the level of customer satisfaction.

Bharill & Rangaraj (2008) study the case of premium services in the passenger segment of Indian Railways. They attempt to derive elasticity estimates between key mode choices internal to the railways and suggest differential pricing to increase revenue on an average. Prasanth & Soman (2009) suggest an RFID technology for ticketing solutions for Indian Railways. They conduct a systematic study of computerization of railways and the loopholes present in the system and propose a simple theoretical model.

Maruvada & Bellamkonda (2010) identify the attributes to evaluate the quality of Railway Passenger Services and develop a comprehensive instrument "RAILQUAL" on the basis of SERVQUAL and Rail Transport quality. Using a fuzzy logic approach to reduce subjectivity and ambiguity of passengers' judgment of service quality, they determine the optimal fuzzy interval of gap scores for each item.

Bogart & Chaudhary (2012) study the relationship between operational costs and state ownership. They report how government ownership of railways reduces the operational costs primarily through cutting labour costs.

Although there have been quite a few attempts to address various issues faced by Indian Railways, we see very limited effort aimed at addressing the problem of locomotive assignment.

5.3 SOUTHERN RAILWAYS

To increase the operational ease, the vast network of Indian Railways is divided in to 16 zones. These zones are further divided in to divisions, each of which has a divisional headquarters. The list of zones and divisions in Indian Railways is detailed in table 11.

These zones manage the operations of trains pertaining to their respective zones. In order to facilitate the operations of trains belonging to these zones, each zone has strategically placed locomotive sheds. Diesel and Electric locomotives are housed in different sheds and at different locations. Table 12 lists the details of different types of electric locomotives housed in electric loco sheds belonging to different zones.

In IR, locomotives are classified by track gauge, motive power, job type and model number. Accordingly, locomotive types are allocated a four or five letter code. The first letter in the code denotes the type of track gauge (broad gauge, meter gauge etc.), the second letter denotes the type of motive power used (electric, diesel etc.), the third letter denotes the type of train the locomotive pulls (goods, passenger, mixed etc.) and the fourth letter denotes the locomotive's model number. The syntax for coding the locomotive type is given below:

- First letter (gauge):
  - W – Broad gauge (wide) – 5 ft 6 in (1,676 mm)
  - Y – Metre gauge (yard) – 3 ft 3 ⅜ in (1,000 mm)
  - Z – 2 ft 6 in (762 mm) narrow gauge
  - N – 2 ft (610 mm) narrow (toy) gauge
- Second letter (motive power):
  - D – Diesel
  - C – DC electric (DC overhead line)
  - A – AC electric (AC overhead line)
  - CA – DC and AC (AC or DC overhead line); CA is considered one letter
  - B – Battery (rare)
- Third letter (job type):
  - G – Goods
  - P – Passenger
  - M – Mixed (goods and passenger)
  - S – Shunting (switching)
  - U – Multiple unit (electric or diesel)
  - R – Railcar

For example, WAP stands for and electric locomotive running of AC power, to be operated on broad gauge for pulling a passenger train.

S/No.	Name of the Railway Zone	Zonal Headquarter	Division
1	Central Railway	Mumbai	1) Mumbai
			2) Nagpur
			3) Bhusawal
			4) Pune
			5) Sholapur
2	Eastern Railway	Kolkata	1) Howrah-I
			2) Howrah-II
			3) Sealdah
			4) Malda
			5) Asansol
			6) Chitaranjan
			7) Kolkata Metro
3	East Central Railway	Hajipur	1) Danapur
			2) Mugalsarai
			3) Dhanbad
			4) Sonpur
			5) Samastipur
4	East Coast Railway	Bhubaneshwar	1) Khurda Road
			2) Waltair
			3) Sambhalpur
5	Northern Railway	Baroda House, New Delhi	1) Delhi-I
			2) Delhi-II
			3) Ambala
			4) Moradabad
			5) Lucknow
			6) Firozpur
6	North Central Railway	Allahabad	1) Allahabad
			2) Jhansi
			3) Agra
7	North Eastern Railway	Gorakhpur	1) Izzatnagar
			2) Lucknow
			3) Varanasi
			4) DLW
8	North Frontier Railway	Maligaon, Guwahati	1) Katihar
			2) Alipurduar
			3) Rangiya
			4) Lumding
			5) Tinsukhia
9	North Western Railway	Jaipur	1) Jaipur
			2) Jodhpur
			3) Bikaner
			4) Ajmer
10	Southern Railway	Chennai	1) Chennai
			2) Madurai
			3) Palghat
			4) Trichy
			5) Trivendrum
11	South Central Railway	Secunderabad	1) Secunderabad
			2) Hyderabad
			3) Guntakal
			4) Vijaywada
			5) Nanded
12	South Eastern Railway	Garden Reach, Kolkata	1) Kharagpur
			2) Adra
			3) Chakradharpur
			4) Ranchi
			5) Shalimar
13	South East Central Railway	Bilaspur	1) Bilaspur
			2) Nagpur
			3) Raipur
14	South Western Railway	Hubli	1) Bangalore
			2) Mysore
			3) Hubli
			4) RWF/YNK
15	Western Railway	Churchgate	1) BCT
			2) Vadodara
			3) Ahemdabad
			4) Ratlam
			5) Rajkot
			6) Bhavnagar
16	West Central Railway	Jabalpur	1) Jabalpur
			2) Bhopal
			3) Kota

Table 11: Zones & Division of Indian Railways (Railways, 2018)

Shed	Rly	WAM 4	WAP 1	WAP 4	WAP 5	WAP 6	WAP 7	WAG 5 A/B	WAG5 HA/HB	WAG 6	WAG 7	WAG 9	WAG 9H	WCAM 1,2,3	WCM 6	WCG 2	WCAG 1	TOTAL
Bhusawal	CR	33		24			92											149
Ajni	CR	1								73	70							144
Kalyan	CR						13			32				53	12	30	2	142
Asansol	ER						35	67		23								125
Howrah	ER	8		84														92
Mughalsarai	ECR	49		40						67								156
Gomoh	ECR					8				79	61							148
Waltair	ECoR	22					107		15	23								167
Angul	ECoR									65								65
Ghaziabad	NR	1	42	41	25	51												160
Ludhiana	NR	1		10				45		78								134
Jhansi	NCR							104		63								167
Kanpur	NCR	3		40						132								175
Arrakonam	SR	30	22					61										113
Erode	SR			80						71								151
Royapuram	SR			46														46
Vijawada	SCR	41					58	71										170
Lallaguda	SCR			60		12				16	66							154
Kazipet	SCR									111								111
TATA	SER	51					14			65	2							132
Bondamunda	SER							31		136								167
Bokaro	SER						52											52
Santragachi	SER			52														52
Bhilai	SECR	36					75			40	7							158
Vadodara	WR	25		57			45	46										173
Valsad	WR							44				47	20					111
Tuglaqabad	WCR	22					74			29	30							155
Etarsi	WCR	52		23			70											145
New Katni Junction	WCR							59		76								135
TOTAL		375	64	557	25	71	635	528	15	1179	236	47	20	53	12	30	2	3849

Table 12: Electric loco holdings in Sheds (Railways, 2016)

Locomotive Type	Top Speed Kmph	HP Rating
WAM4	120	3640
WAP1	130	3800
WAP4	140	5000
WAP5	160	5440
WAP6	160	5060
WAP7	130	6120
WAG5	80	3850
WAG5	80	3850
WAG6	120	6000
WAG7	100	5000
WAG9	100	6120
WAG9	90	6120
WCAM1	120	3640
WCAM2	120	4715
WCAM3	105	5000
WCAG1	100	5000
WCM5	105	4600
WCG2	80	4200

Table 13: Locomotive Ratings (Railways, 2010)

Reliability and safety aspects of electric locomotives are of highest priority which may only be ensured by proper inspections and timely attention. Trip shed is the first line maintenance spot where locomotives are inspected, attended at regular intervals after trips. Regular trip maintenance of locomotives is necessary to avoid deadhauling of locomotives to their home shed in the event of major breakdowns.

Trip sheds are attached to major stations and are generally capable of accommodating two electric locomotives. Trip maintenance of electric locomotives is carried out on completion of 3000 km or one trip; whichever is later and is usually of 2 hours duration. Periodic maintenance of electric locomotives is usually carried out after every 90 days of operation. For periodic maintenance, the locomotives return to the home shed.

The problem we are trying to solve is a single locomotive model which considers the use of only electric locomotives. Since, Southern zone has one of the largest electrified networks and consequently also has one of the largest fleets of electric locomotives, we test the heuristics on the LAP for electric locomotives of SR. For our problem instances, we consider two locomotive sheds viz. Arakkonam and Erode. The types of locomotives we consider for our problem instances are WAM4, WAP1, WAP4, WAP5 and WAP7.

For generating problem instances of larger sizes, we refer to the railway timetable for the south zone. The Southern zone operates numerous trains from which we were able to narrow down the list to 1700 trains in a week which are pulled by electric locomotives. The data pertaining to the type of locomotive pulling a particular train has been obtained from a website managed by the Indian Railways Fan Club. The types of locomotives are updated based on public sightings of trains and locomotives.

The 1700 trains operating on electrified network included trains operating from SR to other zones as well as trains operating within SR. Accordingly, we further narrowed down the list to 420 trains which operate only within SR. We split the list of 420 trains further in to three more problems considering Trivandrum Central, Chennai Central and Chennai Egmore as the three major stations in SR. We also consider an additional problem from the list of 25 test problem instances generated and tested in the earlier chapters by doubling the number of trains but reducing the length of the planning horizon to 16 days. These problem instances include Mail trains, Express trains and Superfast trains.

Mail trains generally have lesser average speed and more stoppages as compared to the other two. Superfast trains on the other hand have the maximum average speed and least number of stoppages among the three. The average speeds and number of stoppages for Express trains lie between the respective figures of those for Mail trains and Superfast trains.

5.3 RESULTS AND DISCUSSIONS

We tested five problem instances of larger sizes as compared to the 25 test problem instances. With respect to solution time, Dijkstra's algorithm-based heuristic outperforms the ACO-based heuristic. The average time taken by Dijkstra's algorithm for solving the five problem instances is 2 hours and 11 minutes with the minimum time recorded at 4 minutes and the maximum time at 6 hours 59 minutes. The corresponding figures for the ACO-based heuristic stand at 2 hours 34 minutes, 27 minutes and, 9 hours and 49 minutes respectively.

In terms of quality of solution, the ACO-based algorithm performs better than the Dijkstra's algorithm-based heuristic in problem instances 2 and 4. For the remaining three problem instances, the Dijkstra's algorithm-based heuristic performs better. The summary of computational tests for Dijkstra's algorithm-based heuristic and ACO-based heuristic are given in table 14. The fleet size for Dijkstra's algorithm-based heuristic and ACO-based heuristic are given in tables 16 and 17 respectively.

Despite the locomotive selection being same for both the heuristic procedures, ACO-based heuristic could at times provide a better solution as compared to the Dijkstra's algorithm-based heuristic. We opine that such a situation could arise when the initial locomotives while trying to serve as many trains as possible, leave out some of the pairing trains unserved. Towards the end of the heuristic, some locomotives might end up serving much lesser number of trains throughout the planning horizon. This effectively leads to a larger fleet size. In contrast, had the pairing trains not been left out, the number of locomotives could at times (not necessarily always) lead to smaller fleet size. Since, ACO-based heuristic decides the next path based on relative probability, there always exists a possibility of all pairing trains being clubbed together to be served by one locomotive and consequently arrive at a better solution than Dijkstra's algorithm-based heuristic.

The reasons for such occurrences need to be further investigated which could form part of future studies. A t-Test to compare the results of Dijsktra's Algorithm-based Heuristic and ACO-based Heuristic shows that there exists no statistically significant difference between the results of the two algorithms.

No. of loco types	No. of stations	Planning horizon	No. of Trains	Time gap between two consecutive time slots	Termination time HH:MM	
					Dijkstra’s Algorithm based Heuristic	ACO-based Heuristic
5	14	16 days	84	0.5	2:59	1:23
5	8	16 days	164	4	0:26	0:41
5	9	16 days	104	4	0:04	0:32
5	8	16 days	122	4	0:27	0:27
5	22	16 days	420	4	6:59	9:49

Table 14: Summary of Computational Tests for SR

Paired Samples Statistics

		Mean	N	Std. Deviation	Std. Error Mean
Pair 1	D	131.00	5	175.512	78.491
	ACO	154.40	5	243.950	109.098

Paired Samples Correlations

		N	Correlation	Sig.
Pair 1	D & ACO	5	.948	.014

Paired Samples Test

		Paired Differences					t	df	Sig. (2-tailed)
		Mean	Std. Deviation	Std. Error Mean	95% Confidence Interval of the Difference				
					Lower	Upper			
Pair 1	D - ACO	-23.400	95.377	42.654	-141.826	95.026	-.549	4	.612

Table 15: t-Test for Dijsktra's Algorithm based Heuristic vs ACO-based Heuristic

Shed 1					Shed 2					Total
WAM4	**WAP1**	**WAP4**	**WAP5**	**WAP7**	**WAM4**	**WAP1**	**WAP4**	**WAP5**	**WAP7**	
5	0	0	0	0	2	1	0	0	0	8
0	0	18	0	0	0	0	20	19	0	57
0	0	0	0	0	7	0	15	0	0	22
10	0	0	0	0	20	0	14	0	0	44
20	20	0	0	8	20	20	0	0	20	108

Table 16: Fleet size for SR - Dijkstra's algorithm-based Heuristic

Shed 1					Shed 2					Total
WAM4	**WAP1**	**WAP4**	**WAP5**	**WAP7**	**WAM4**	**WAP1**	**WAP4**	**WAP5**	**WAP7**	
5	0	0	0	0	2	2	0	0	0	9
0	0	13	6	0	0	0	17	15	0	51
3	0	5	0	0	9	1	16	0	0	34
7	0	0	0	0	20	3	6	0	0	36
20	15	6	2	7	20	20	8	5	8	111

Table 17: Fleet size for SR - ACO-based Heuristic